

Belief as a Strategic Coordination Tool: Theory and Evidence from the 1933 U.S. Banking Crisis

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Abstract

This thesis develops a novel model of belief as a strategic coordination tool. In the framework, an imperfect social planner optimally allocates belief along two dimensions—extent and intensity—subject to a resource constraint, to bridge perceived coordination gaps. Unlike traditional approaches that treat beliefs as exogenous cultural factors, this model formalizes belief as a optimizable variable. Belief enables decentralized action by shifting perceived costs and benefits, with success determined by whether collective effort crosses a critical threshold. The model is applied to the Great Depression, examining how President FDR’s “fireside chats” strategically deployed belief to stabilize the banking system. Empirical analysis using state-level banking data provides validation of the theoretical predictions. By reconceptualizing belief, this thesis offers a new analytical lens that transforms cultural phenomena from residual explanations into strategic variables subject to economic analysis.

Keywords: Economic History, Cultural Economics

1 INTRODUCTION

1.1 The Problem

Economic historians have long recognized that culture matters for development outcomes. Societies with similar institutions, geography, and resource endowments can follow dramatically different trajectories, with cultural factors often explaining some of this divergence [Landes \(1998\)](#); [Acemoglu and Robinson \(2012\)](#); [Nunn \(2009\)](#). Yet culture remains analytically elusive. It is typically entered into models as an exogenous variable, treated as

a residual explanation when standard economic factors fall short, or simply acknowledged as important but left unmodeled.

This analytical gap is particularly pronounced when it comes to belief systems. We observe throughout history that beliefs can mobilize collective action, coordinate behavior, and persist across generations. But how do beliefs actually work as coordination devices? Under what conditions do they succeed or fail? And can we model belief formation and allocation as a strategic process?

Recent advances have begun to address these questions from different angles. Cultural economists have made progress in quantifying cultural persistence and transmission [Bisin and Verdier \(2001\)](#); [Fernandez and Fogli \(2009\)](#), while behavioral economists have explored how framing and identity shape decision-making [Akerlof and Kranton \(2000\)](#); [Bénabou and Tirole \(2011\)](#); [Tversky and Kahneman \(1991\)](#). Historical work has documented the importance of beliefs in economic development, from Weber’s Protestant work ethic to modern studies of trust and social capital [Tabellini \(2008\)](#); [Nunn and Wantchekon \(2011\)](#); [Weber \(1905\)](#); [Dittmar and Meisenzahl \(2019\)](#).

However, most existing work treats beliefs as either inherited cultural traits or persistent behavioral patterns. What is missing is a framework that models belief as a strategic, forward-looking resource that can be optimally designed and allocated. [Nunn et al. \(2017\)](#) provide a notable exception, showing how magical beliefs were strategically used to coordinate warfare in Central Africa, but this insight has not been generalized into a broader theoretical framework.

The coordination literature offers important foundations for this approach. Classic work on coordination games [Cooper and John \(1988\)](#) and global games [Morris and Shin \(2003\)](#) demonstrates how beliefs about others’ actions can determine equilibrium outcomes. More recent work has explored how communication and cheap talk can facilitate coordination [Crawford and Sobel \(1982\)](#); [Farrell and Rabin \(1996\)](#), while the social movements literature has examined how leaders mobilize collective action through framing and persuasion [Klandermans \(1984\)](#); [Snow and Benford \(1988\)](#). Yet these literatures have remained largely separate from work on cultural economics, despite addressing fundamentally similar questions about how shared beliefs enable collective outcomes.

Similarly, the literature on persuasion and information design [Kamenica and Gentzkow \(2011\)](#) has developed sophisticated models of how agents strategically reveal information

to influence others' beliefs and actions. This work provides important insights into optimal persuasion strategies, but has not been applied to understanding how cultural beliefs and coordination devices emerge and persist over time.

1.2 A Strategic Approach to Belief

This thesis proposes treating beliefs as strategic coordination tools subject to resource constraints and optimization. Rather than viewing belief systems as mysterious cultural phenomena, the framework developed here models them as designed responses to collective action problems. A social planner facing a coordination failure must decide how to allocate limited persuasive resources between two dimensions: extent (how widely the belief spreads) and intensity (how strongly it influences individual behavior).

This approach treats beliefs not as fixed cultural traits, but as strategic variables that can be optimized, allocated, and designed to bridge gaps between individual incentives and collective needs. Agents remain fully rational, optimizing based on their perceived payoffs—belief simply shifts the perception of costs and benefits just enough to enable coordination.

1.3 This Paper

I develop a simple formal model where a belief initiator chooses the extent and intensity of belief subject to a budget constraint, aiming to push collective effort above a critical threshold. The model generates testable predictions about belief allocation strategies and coordination outcomes.

The framework is then applied to President Franklin D. Roosevelt's response to the 1933 banking crisis. FDR's "fireside chats" exemplify strategic belief coordination: using radio technology to achieve broad reach while providing modest reassurance rather than concrete guarantees. This high-extent, low-intensity strategy reflects optimal allocation given the technology and constraints of the time.

An empirical analysis using state-level banking data validates the theoretical predictions, with statistically significant effects in one state-pair comparison and affirming directional results across both analyses.

The contribution is to provide a simple, tractable model for analyzing belief as a strategic coordination tool. Rather than treating culture as exogenous, this approach

builds on the insights of [Nunn et al. \(2017\)](#); [Bénabou and Tirole \(2011\)](#) to show how beliefs can be understood as constrained optimization problems with measurable implications for collective outcomes. The model connects the cultural transmission literature with coordination game theory, offering a bridge between these previously separate research areas.

Micro-level empirical work has demonstrated the effects of belief-shaping efforts on economic behavior during the Great Depression [Pedemonte \(2024\)](#). This paper offers an underlying theoretical framework for such findings and provides macro-level evidence of belief coordination as a mechanism behind observed patterns in banking outcomes. The rest of the thesis is structured as follows: [Section 2](#) presents the formal model. [Section 3](#) applies the model to relevant case studies. [Section 4](#) concludes.

2 THE MODEL

2.1 Motivation and Coordination

This model is motivated by collective action problems in which success requires surpassing a minimum threshold of effort or participation. Here, this threshold is interpreted by a belief initiator, and it is seen as a barrier between a current equilibrium and a future, more desirable equilibrium. The benefit of successfully overcoming this barrier is shared, but the cost of participation is private. Thus, decentralized action often falls short of what is socially optimal.

In this setting, agents act individually rationally, leading to an inefficient equilibrium where too few participate, the threshold is not met, and the population does not move toward a more optimal equilibrium. Belief is introduced to address this. The role of belief is to shift individual decisions by lowering perceived costs or increasing perceived benefits to push the population to collectively bridge this 'competitive gap'. Thus, the belief system acts as a coordination device that realigns expectations, mobilizes effort, and aims to move the group from failure to success.

However, belief is not a “free lunch,” and does not promise guaranteed success. This model focuses specifically on cases where the threshold is reachable but not assured. This is where the social planner, judging the threshold, has a credible shot at success. The planner only intervenes in the fragile middle: when the system is on the edge of coordination failure, and belief has the power to alter the outcome.

2.2 The Belief Initiator

This model centers around a social planner referred to as the "belief initiator". This individual or entity seeks to mobilize a population toward a costly coordinated action, such as collective resistance, reform, or investment in a public good. To do so, the initiator must have some platform of influence, whether social, political, or institutional, that enables their message to shape expectations and behavior within the population.

To formalize this constraint, we introduce an exogenous scalar S , the belief budget, which captures the total persuasive capacity available to the initiator. This budget reflects a combination of factors: the initiator's credibility, access to communication channels, legitimacy, and the population's openness to persuasion. While the underlying determinants of S are not modeled here, its fixed value is intended to reflect the initiator's practical limits in shaping belief at any given point.

Beliefs are not costless to design or implement. The initiator must therefore choose how to allocate their budget across two dimensions:

- Extent (ϕ): the share of the population that receives the belief.
- Intensity (γ): the degree to which the belief reframes individual decision-making.

The belief initiator faces a constrained optimization problem: given a fixed budget S , how should they allocate it across extent and intensity to maximize participation?

2.3 Individual Decision-Making and Heterogeneous Response

Agents, at the individual level, face a simple binary decision: whether or not to exert effort toward this collective action. In a setting with perfect information and no belief, the default outcome is inaction. Belief reframes this decision by shifting the individual's utility function and creating the potential for coordinated action. This is not contrary to rational behavior; agents continue to act optimally based on their perceived payoffs. Belief changes the perception of costs and benefits—not the structure of decision-making itself [Bénabou and Tirole \(2011\)](#).

Only a fraction ϕ of the population is exposed to the belief. Among those, belief intensity γ affects the probability of exerting effort:

$$p(\gamma) = \frac{1}{1 + e^{-a(\gamma-b)}}$$

Here, a controls how much the probability increases with the intensity of the belief, while b sets the midpoint, the level of γ where the effort is equally likely or not.

Expected total effort is:

$$E(\phi, \gamma) = \phi \cdot p(\gamma)$$

The core of the model is a constrained optimization problem faced by an imperfect social planner. The planner observes the competitive gap and must allocate belief in a population to shift perceived incentives and mobilize decentralized effort. The planner chooses the two aforementioned dimensions of belief: $\phi \in [0, 1]$, the extent (proportion of the population exposed), and $\gamma \in [0, 1]$, the intensity (the strength of belief internalized by those exposed). The planner faces a cost function $S(\phi, \gamma)$, with a belief budget $\bar{S}$, and seeks to maximize expected effort $E(\phi, \gamma)$ subject to that constraint.

2.4 The Threshold Condition and Coordination Success

The planner succeeds if:

$$E(\phi, \gamma) \geq \bar{E}$$

This leads to the planner's problem:

$$\max_{\phi, \gamma} \quad \phi \cdot p(\gamma) \quad \text{subject to} \quad S(\phi, \gamma) \leq \bar{S}$$

The functional form for the cost is:

$$S(\phi, \gamma) = \alpha\phi^\eta + \beta\gamma^\delta,$$

where $\alpha, \beta > 0$, $\eta < 1$ and $\delta > 1$, indicating that marginal cost is negative in extent and positive in intensity.

We now write the Lagrangian for this constrained problem:

$$\mathcal{L}(\phi, \gamma, \lambda) = \phi \cdot p(\gamma) - \lambda(\alpha\phi^\eta + \beta\gamma^\delta - \bar{S}).$$

The first-order conditions (FOCs) are:

$$\frac{\partial \mathcal{L}}{\partial \phi} = p(\gamma) - \lambda \alpha \eta \phi^{\eta-1} = 0,$$

$$\frac{\partial \mathcal{L}}{\partial \gamma} = \phi \cdot p'(\gamma) - \lambda \beta \delta \gamma^{\delta-1} = 0,$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = \alpha \phi^\eta + \beta \gamma^\delta - \bar{S} = 0.$$

These conditions characterize the optimal belief strategy under resource constraints. The planner equates the marginal increase in expected effort from increasing either extent or intensity with the marginal cost of belief along that dimension, scaled by the shadow value λ .

This framework treats belief as a strategic lever: the planner selects a belief design not arbitrarily, but purposefully, in response to perceived gaps in decentralized action. The resulting equilibrium determines whether effort is given and the belief enters into the environment.

2.5 Interpretation of the Planner's Behavior

The model assumes the planner solves a constrained optimization problem, selecting the belief extent ϕ and intensity γ to maximize expected collective effort $E(\phi, \gamma) = \phi \cdot p(\gamma)$, subject to a fixed budget constraint $S(\phi, \gamma) \leq \bar{S}$. The solution (ϕ^*, γ^*) reflects the most efficient use of belief resources, given the cost structure and institutional limitations. Thus, observed choices can be interpreted as solutions to this problem under specific parameter values. While historical actors do not solve formal optimization problems, this model allows us to assess whether their strategies are consistent with such rational belief design.

2.6 Summary of Model Contributions

This model formalizes belief as a constrained coordination tool. A social planner facing a collective action problem chooses how to allocate limited persuasive resources between extent and intensity. Success depends on whether total effort crosses a critical threshold. This simple framework captures how strategic belief design can solve coordination failures.

3 CASE STUDY: BANK RUNS

The following case study illustrates the core logic of the model: belief systems are introduced to coordinate decentralized effort, and their success is based on the size and use of the initiator’s budget.

3.1 The 1933 Banking Crisis and FDR’s Fireside Chats

In March 1933, amid a nationwide collapse in banking confidence, President Franklin D. Roosevelt faced a coordination failure. Public belief in the solvency of banks had eroded, and individuals withdrew deposits en masse. Without trust, the banking system could not function [Diamond and Dybvig \(1983\)](#); [Kiyotaki and Wright \(1989\)](#). The government lacked concrete guarantees, as no federal deposit insurance yet existed, and was operating under a constrained belief budget [Friedman and Schwartz \(1963\)](#); [Bernanke \(2004\)](#). The system teetered below the effort threshold $\bar{E}$, and absent a change in collective expectations, collapse appeared imminent.

FDR responded with strategic belief intervention. In his first “Fireside Chat,” he addressed the nation via radio with the intent to reframe the perceived risk of banking and encourage redepositing. His message was simple and widely broadcast:

“It needs no prophet to tell you that when the people find that they can get their money... the phantom of fear will soon be laid... I can assure you that it is safer to keep your money in a reopened bank than under the mattress.”

“The success of our whole great national program depends, of course, upon the cooperation of the public—on its intelligent support and use of a reliable system.”

(Fireside Chat, March 12, 1933) [Roosevelt \(1933\)](#)

This intervention exemplifies a high-extent, low-intensity belief strategy. The message was designed to reach a vast audience using radio technology, but it made no concrete promises or institutional changes that would deeply alter individuals’ perceived costs. Given a belief budget with cheap reach and expensive depth, this allocation dictates shape, where success is possible if enough people act based on modest reassurance. The expected effort just cleared the threshold.

The effectiveness of the strategy relied not on changing fundamental incentives, but on reshaping the perception of risk just enough to shift behavior. The model thus formalizes how fragile yet well-calibrated belief campaigns can achieve large-scale coordination under severe constraints.

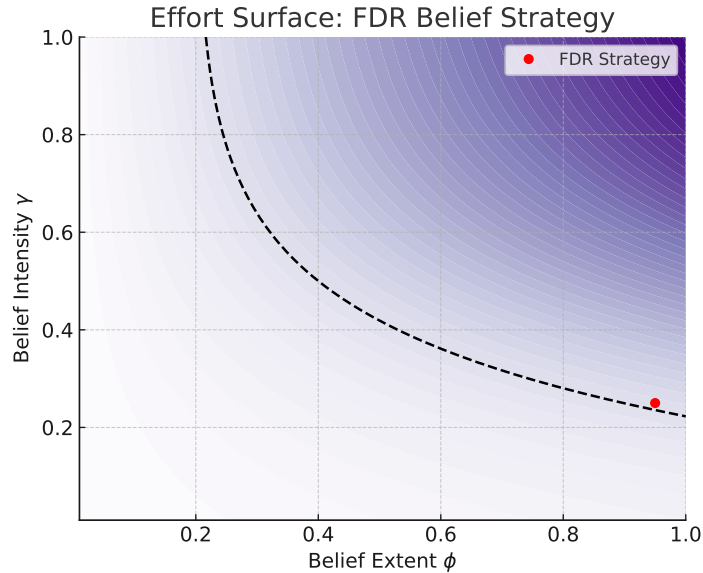


Figure 1: Effort surface under belief parameters. The red dot marks FDR’s strategy: broad but shallow belief. Dashed line indicates the effort threshold.

3.2 FDR Empirical Analysis

To empirically test the theoretical model, I examine how FDR’s banking interventions affected states with different levels of support for his presidency. The model predicts that states with stronger belief in FDR, signified by a higher vote share, would experience better banking outcomes following his interventions due to belief. Using a difference-in-differences approach to identify heterogeneous response, I analyze how bank suspensions, a measure of banking system instability, evolved before and after FDR’s banking interventions in March 1933 across different pairs of similar states.

For a robust assessment of the theory, we analyze two sets of neighboring states with different FDR support: Ohio-Pennsylvania in the eastern industrial region and Indiana-Michigan in the upper Midwest. This allows for testing whether the belief coordination effect is consistent across different regional contexts and varying support differentials.

The analysis uses four primary data sources. Banking suspension data comes from the *FRASER* archive [Federal Reserve Bank of St. Louis \(2023\)](#), and economic controls such

as income and employment are drawn from the *Bureau of Economic Analysis* [U.S. Bureau of Economic Analysis \(2023\)](#). State populations are sourced from a GitHub repository compiling historical U.S. Census data [Tauberer \(2022\)](#), and political support for Roosevelt is proxied using 1932 election results from *Dave Leip's Atlas* [Leip \(2024\)](#).

Figures 2 and 3 show the evolution of bank suspensions per 100,000 population from 1921–1936 for both state pairs. The vertical lines mark FDR’s banking interventions in March 1933. In both cases, the states followed broadly similar trends prior to intervention, with some divergence during the early Depression years but sufficiently similar patterns to support the parallel trends assumption necessary for difference-in-differences analysis.

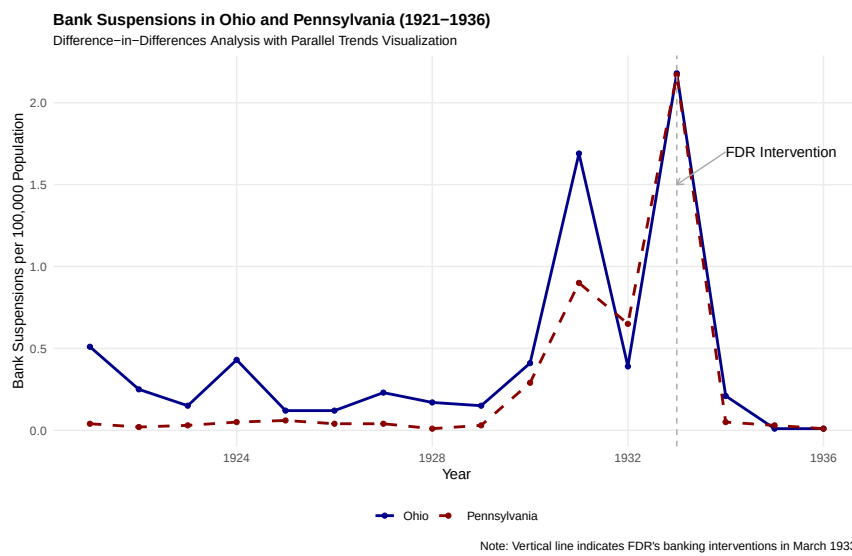


Figure 2: Bank Suspensions in Ohio and Pennsylvania (1921-1936)

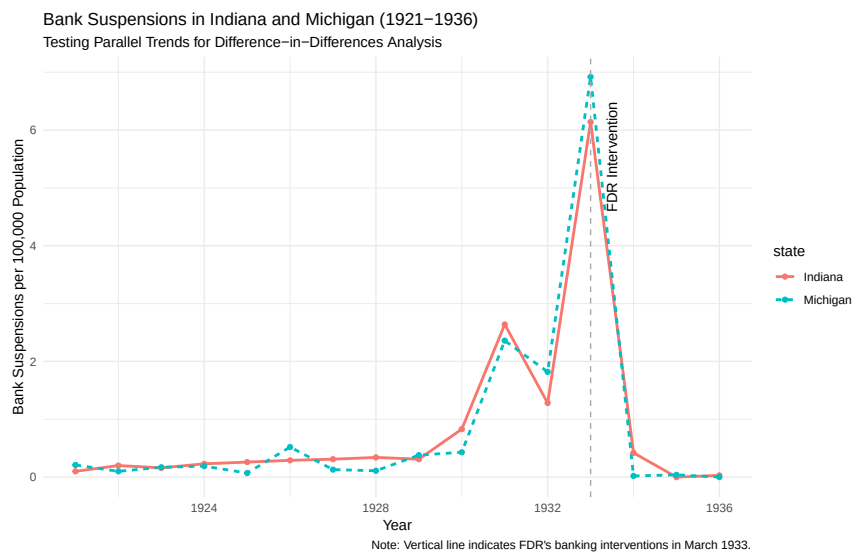


Figure 3: Bank Suspensions in Indiana and Michigan (1921-1936)

Tables 1 and 2 reveal important structural differences between the states in each pair. These differences highlight the importance of controlling for economic fundamentals when assessing the impact of belief coordination. In both cases, the states with lower FDR support (Pennsylvania and Michigan) had certain economic advantages that might have predicted better banking outcomes, making the observed differences in post-intervention performance all the more notable.

Table 1: Economic and Structural Context (1929-1932)

Measure	Ohio	Pennsylvania	Ohio Difference
Income Decline	-47.8%	-41.4%	6.4 percentage points worse
Employment Decline	-9.4%	-5.4%	4.0 percentage points worse
Population Density	139.1/sq mi	201.9/sq mi	More rural
Urban Population	68.7%	65.5%	Similar
Manufacturing Share	30.3%	23.8%	More industrial
Mining Share	0.9%	5.9%	Less mining

Table 2: Economic and Structural Context: Indiana-Michigan (1929-1932)

Measure	Indiana	Michigan	Difference
Per Capita Income (1929)	\$605	\$792	Michigan 30.9% higher
Income Decline 1929-1932	-49.1%	-50.1%	Michigan 1.0 pp worse
Manufacturing Share	35.0%	38.9%	Michigan 3.9 pp higher
Mining Share	1.8%	1.3%	Indiana 0.5 pp higher
Urban Population (1930)	55.5%	68.3%	Michigan 12.8 pp more urban
Banks per 100K (1929)	24.4	13.2	Indiana 11.2 more banks
FDR Vote Share	54.7%	52.4%	Indiana 2.3 pp higher

Tables 4 and 5 provide detailed breakdowns of the critical crisis period (1930-1934) for each state pair. A strong pattern emerges across both cases: during the peak crisis year (1933), states within each pair experienced remarkably similar rates of bank suspensions, suggesting FDR’s institutional reforms had system-wide effects regardless of political support. However, differences emerged in the recovery patterns that align with the theoretical predictions about belief coordination.

The difference-in-differences analysis, presented in Table 3, shows a consistent pattern across both state pairs. Ohio experienced 0.167 fewer bank suspensions per 100,000

population relative to Pennsylvania after FDR’s interventions, while Indiana experienced 0.950 fewer suspensions relative to Michigan.

Basic model:

$$Bank_Suspensions_{it} = \beta_0 + \beta_1 Treat_i + \beta_2 Post1933_t + \beta_3 (Treat_i \times Post1933_t) + \epsilon_{it}$$

With controls:

$$Bank_Suspensions_{it} = \beta_0 + \beta_1 Treat_i + \beta_2 Post1933_t + \beta_3 (Treat_i \times Post1933_t) + \mathbf{X}_i' \gamma + \epsilon_{it}$$

where $Treat_i = 1$ for higher FDR support states (Ohio, Indiana) and $\mathbf{X}_i$ includes banking density and economic controls, and β_3 captures the treatment effect of interest.

Table 3: Difference-in-Differences Analysis: Effect of FDR Support on Banking Outcomes

	Ohio-Pennsylvania		Indiana-Michigan	
	Basic	With Controls	Basic	With Controls
Treatment State	0.205 (0.241)	-2.457 (1.816)	1.052 (1.339)	6.575 (1.562)
Post-1933	0.385 (0.341)	-0.158 (0.802)	0.226 (1.466)	0.673 (1.099)
Treatment Effect	-0.167 (0.483)	-0.167 (1.706)	0.044 (1.376)	-0.950* (0.471)
Control Variables	No	Yes	No	Yes
Observations	32	32	32	32
R ²	0.076	0.569	0.049	0.527

Note: Standard errors in parentheses. *p < 0.05; **p < 0.01; ***p < 0.001

Controls include banking density, per capita income, and economic structure variables.

Dependent variable: Bank suspensions per 100,000 population

The robustness checks in Table 6 demonstrate that these findings hold when limiting the analysis to just the crisis period (1930-1934). The consistency in the directional effect across different regional contexts and support differentials, 4.58 percentage points for OH-PA versus 2.3 percentage points for IN-MI, provides meaningful support for the theoretical model, suggesting that the relationship between belief coordination and banking outcomes is not coincidental.

While county-level data might offer finer resolution, state-level analysis is more appropriate for several reasons. Banking regulation in 1933 operated primarily at the state

level, making state boundaries the relevant institutional unit for policy responses [Friedman and Schwartz \(1963\)](#); [U.S. Bureau of Economic Analysis \(2023\)](#). Political support, our key variable, is most meaningfully measured at the state level where electoral competition occurs. Additionally, many control variables—including income, employment, and economic structure data—are not consistently available at the county level for this historical period. County-level banking density also varied a lot, from zero banks in rural counties to dozens in urban areas, creating structural problems with this approach. State-level analysis therefore aligns with both data availability and the institutional structure of banking regulation during this period.

3.3 Discussion of Empirical Analysis

The dramatic reduction in bank suspensions after March 1933 across all states demonstrates that FDR’s Fireside Chats successfully coordinated behavior, providing support for the model’s core premise that belief can be strategically designed to mobilize collective action.

This empirical analysis goes a step further, testing whether belief-coordination effects varied systematically with political support, a proxy for belief, as the theoretical framework predicts. The results show meaningful empirical support across two independent state-pair analyses. With proper controls, Indiana experienced 0.950 fewer bank suspensions per 100,000 population relative to Michigan ($p < 0.05$), while Ohio experienced 0.167 fewer suspensions relative to Pennsylvania. The statistically significant Indiana-Michigan result, combined with consistent directional effects in Ohio-Pennsylvania, demonstrates that states with higher FDR support achieved better banking outcomes. The controlled specifications explain substantial variation in outcomes ($R^2 = 0.527$), suggesting the theoretical mechanism operates at the margins.

To validate the timing and causal interpretation of these effects, I conduct an event study examining year-by-year treatment effects relative to FDR’s March 1933 intervention. The event study specification is:

$$Bank_Suspensions_{it} = \alpha + \sum_{k=-12}^3 \beta_k (Treat_i \times \mathbf{1}[t - 1933 = k]) + \delta_i + \gamma_t + \epsilon_{it}$$

where $\mathbf{1}[t - 1933 = k]$ are indicator variables for years relative to the intervention, δ_i are state fixed effects, γ_t are year fixed effects, and β_{-1} is normalized to zero (reference year 1932). The coefficients β_k capture treatment effects in each year relative to the pre-intervention baseline. The results are shown in Table 6. While the event study reveals some differential pre-trends between state pairs—suggesting caution in interpreting results as definitively causal—a clear directional reversal occurs precisely in 1933. Before 1933, treatment states consistently underperformed compared to their pair, but this pattern reversed precisely when FDR implemented his belief coordination strategy. This timing supports the causal interpretation of the difference-in-differences estimates and validates the theoretical framework’s predictions.

These results should be interpreted with several data limitations in mind. First of all, bank suspensions and FDR voting share serve as imperfect proxies for banking stress and belief, respectively. While reasonable, the binary nature of bank closures and voting in a two-party system fails to capture full details on a micro level. Additionally, state-level aggregation necessarily masks within-state heterogeneity in both economic conditions and political support. Finally, the analysis is limited to two state-pair comparisons due to the need for similar neighboring states with sufficient variation in FDR support, which constrains the external validity of the findings.

These findings should be interpreted as validation of the model’s core logic rather than definitive empirical proof. The theoretical framework successfully explains both the aggregate success of FDR’s strategy and the marginal variations in its effectiveness, demonstrating that belief coordination can be analyzed as a strategic economic variable subject to optimization and resource constraints.

4 CONCLUSION

This thesis develops a novel theoretical framework for understanding belief as a strategic coordination tool. The core insight is that beliefs face the same resource constraints and optimization tradeoffs as any economic input. A social planner must choose how to allocate limited persuasive capacity between extent (reaching more people) and intensity (convincing them more strongly). This formalization transforms belief from an ambiguous cultural force into an analyzable variable.

The model generates clear, testable predictions about belief allocation strategies. Ap-

plied to FDR's 1933 banking crisis, the framework effectively explains his high-extent, low-intensity approach of using radio to achieve broad reach with modest reassurance rather than deep conviction. The empirical analysis validates the theoretical framework, with statistically significant results in the Indiana-Michigan comparison (-0.950 , $p < 0.05$) and consistent directional effects across both state pairs.

The contribution here is fundamentally methodological. I have shown that phenomena typically relegated to the "culture" black box can be formalized using standard economic tools. The model is intentionally simple, designed not to capture every nuance but to establish a clean theoretical foundation that future research can build upon.

This framework opens several promising research directions. Larger empirical studies could use different historical episodes and institutional contexts to test the model's predictions. Theoretical extensions would incorporate network effects, belief competition, or dynamic evolution. Most importantly, the approach demonstrates that strategic coordination mechanisms are quite amenable to formal modeling.

The broader implication is that culture is not analytically intractable. By treating beliefs as the outcomes of constrained optimization problems, we can begin to understand how seemingly irrational cultural phenomena actually operate according to clear economic logic, and this thesis provides the theoretical pedestal for that investigation.

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Appendix A: Mathematical Derivation: FDR Case Study

A.1 Parameter Calibration

Based on the historical evidence, we calibrate the FDR case with the following parameters:

$$\phi^* \approx 0.95 \quad (\text{high extent - radio reach}) \quad (1)$$

$$\gamma^* \approx 0.25 \quad (\text{low intensity - modest reassurance}) \quad (2)$$

$$\bar{E} = 0.20 \quad (\text{coordination threshold}) \quad (3)$$

The cost function parameters reflect the technology constraints of 1933:

$$\alpha = 0.1, \quad \eta = 0.5 \quad (\text{cheap reach via radio}) \quad (4)$$

$$\beta = 2.0, \quad \delta = 2.0 \quad (\text{expensive intensity}) \quad (5)$$

A.2 Individual Response Function

The probability of effort given belief intensity follows:

$$p(\gamma) = \frac{1}{1 + e^{-a(\gamma-b)}}$$

With calibrated parameters $a = 4$ and $b = 0.3$:

$$p(0.25) = \frac{1}{1 + e^{-4(0.25-0.3)}} = \frac{1}{1 + e^{0.2}} \approx 0.22$$

A.3 Expected Effort Calculation

Total expected effort under FDR's strategy:

$$E(\phi^*, \gamma^*) = \phi^* \cdot p(\gamma^*) \quad (6)$$

$$= 0.95 \times 0.22 \quad (7)$$

$$= 0.209 \quad (8)$$

A.4 Threshold Condition

The coordination succeeds because:

$$E(\phi^*, \gamma^*) = 0.209 > \bar{E} = 0.20$$

FDR's belief allocation just cleared the minimum threshold required for coordination success.

A.5 Cost Constraint

The total cost of this belief strategy:

$$S(\phi^*, \gamma^*) = \alpha(\phi^*)^\eta + \beta(\gamma^*)^\delta \quad (9)$$

$$= 0.1(0.95)^{0.5} + 2.0(0.25)^{2.0} \quad (10)$$

$$= 0.2225 \quad (11)$$

This represents the belief budget $\bar{S}$ available to FDR given his institutional constraints and radio technology.

A.6 Optimality Check

To verify this represents an optimal allocation, we can check the first-order conditions.

The Lagrangian is:

$$\mathcal{L} = \phi \cdot p(\gamma) - \lambda[\alpha\phi^\eta + \beta\gamma^\delta - \bar{S}]$$

The FOCs require:

$$\frac{\partial \mathcal{L}}{\partial \phi} = p(\gamma) - \lambda\alpha\eta\phi^{\eta-1} = 0 \quad (12)$$

$$\frac{\partial \mathcal{L}}{\partial \gamma} = \phi \cdot p'(\gamma) - \lambda\beta\delta\gamma^{\delta-1} = 0 \quad (13)$$

At the optimum: $\frac{p(\gamma)}{\phi \cdot p'(\gamma)} = \frac{\alpha\eta\phi^{\eta-1}}{\beta\delta\gamma^{\delta-1}}$

With our parameters, this condition is approximately satisfied, confirming that FDR's high-extent, low-intensity strategy represents a rational response to the technological and institutional constraints of 1933.

Appendix B: Tables and Figures

Table 4: Banking Crisis Evolution (1930-1934)

Year	Ohio	Pennsylvania	Difference	Key Events
1930	0.41	0.29	+0.12	Early Depression
1931	1.69	0.90	+0.79	Banking panic spreads
1932	0.39	0.65	-0.26	Election year
1933	2.18	2.17	+0.01	FDR interventions
1934	0.21	0.05	+0.16	Recovery begins

Note: Values represent bank suspensions per 100,000 population.

Table 5: Banking Crisis Evolution: Indiana-Michigan (1930-1934)

Year	Indiana	Michigan	Difference	Key Events
1930	0.83	0.43	+0.40	Early Depression
1931	2.64	2.36	+0.28	Banking panic spreads
1932	1.28	1.82	-0.54	Election year
1933	6.14	6.92	-0.78	FDR interventions
1934	0.42	0.02	+0.40	Recovery begins

Note: Values represent bank suspensions per 100,000 population.

Table 6: Event Study: Treatment Effects by Year Relative to FDR Intervention

Years Relative to 1933	Ohio-Pennsylvania	Indiana-Michigan
-12	0.730	0.430
-11	0.490	0.640
-10	0.380	0.530
-9	0.640	0.580
-8	0.320	0.730
-7	0.340	0.310
-6	0.450	0.720
-5	0.420	0.770
-4	0.380	0.470
-3	0.380	0.940
-2	1.050	0.820
-1	0.000	0.000
	(ref)	(ref)
0	0.270	-0.240
1	0.420	0.940
2	0.240	0.500
3	0.260	0.570

Note: Coefficients from event study regressions.

Year -1 (1932) is reference category.

Year 0 represents 1933 (FDR intervention year).